Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

North West European Shelf Production Centre
NWSHELF_ANALYSISFORECAST_BGC_004_002

Issue: 6.0

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Approval date by the CMEMS product quality coordination team: 23/10/2025

CHANGE RECORD

When the quality of the products changes, the QuID is updated, and a row is added to this table.

The third column specifies which sections or sub-sections have been updated. The fourth column should mention the reason for updating the document. Column 5 should include the names of the author(s) doing the changes. Including the name in column 6 shows that the document has been validated by the TAC/MFC PQ leader or deputy (or TAC/MFC PQ leader or the deputy). This validation means the document was checked for contents, formats and consistency with other documents.

Issue	Date	§	Description of Change	Author	Validated By
1.0	16/09/2011	All	First version of document		
2.0	28/10/2011	All	Updated version of document		
2.1	18/11/2011	V, VI	Minor corrections	A. Sellar	
3.0	15/12/2014	All	New version for MyOcean V5.0 product upgrade	A. McLaren et al.	Ed Blockley
3.1	26/02/2015	All	Revision after V5 acceptance	A. McLaren et al.	Ed Blockley
3.2	May 1 2015	All	Change format to fit CMEMS graphical rules		L. Crosnier
3.3	20/01/2016	I.2, I.3, III.3, IV.6-IV.7	Revision for V2 (MLD and bottom temperature validation)	P. Sykes	M. Tonani
3.3	01/04/2016	I.3, IV.4, IV.6, VI	Revision after CMEMS V2 Acceptance	P. Sykes and M. Tonani	M. Tonani
3.4	6/01/2017		NORTHWESTSHELF_ANALYSIS_FORECAST –PHYS-004-001_b moved to a different document..	M. Tonani	M. Tonani
4.0	10/09/2019	All	Revision for V10	S. Kay	M. Tonani
4.1	10/09/2020	All	Revision for V11	R.McEwan	M. Tonani
4.2	07/01/2021	All	Product name updated	I. Ascione	M. Tonani
5.0	31/07/2023	All	**[EIS November 2023]** Complete system update implying change of the modelling system configuration and resolution.	E. Gutknecht G. Reffray S. Ciliberti	S. Cailleau
5.1	31/08/2024		**[EIS November 2024: IBI-V7 version]** New variable distributed: diffusive attenuation coefficient of light in sea water (Kd)	Elodie Gutknecht, Guillaurme Reffray, Stefania Ciliberti	Sylvain Cailleau

Issue	Date	§	Description of Change	Author	Validated By
6.0	28/05/2025	All	**[EIS November 2025]** Complete system update	S. Kay	T.Brüning

ACRONYM TABLE

Acronym	Description
CMEMS	Copernicus Marine Environment Monitoring Service
DIC	Dissolved Inorganic Carbon
EAN	Estimated Accuracy Number
ERSEM	European Regional Seas Ecosystem Model
GLODAP	Global Ocean Data Analysis Project
ICES	International Council for the Exploration of the Sea
MY	Multi-year
NRT	Near Real Time
NWS	North West [European] Shelf
pCO2	Partial pressure of CO2
RMSD	Root Mean Square Difference
SOCAT	Surface Ocean CO_2 Atlas
TAC	Copernicus Marine Service Thematic Assembly Centre

KEY DEFINITIONS

NWS shelf region (definition according to Wakelin et al. 2012)	The region mask of Wakelin et al. (2012) is used to divide the model domain into 13 regions (see Figure 2). The NRT model domain includes the North West European shelf seas, as well as other regions such as part of the North East Atlantic Ocean, the Norwegian Trench, the Skagerrak and Kattegat. When we refer to the North West Shelf region, we only refer to a portion of the model domain. The North West Shelf region is defined by combining a number of regions from the Wakelin et al. (2012) region mask. These regions are: the Southern North Sea; Central North Sea; Northern North Sea; English Channel; Shetland Shelf; Irish Shelf; Irish Sea; and the Celtic Sea region.
Climatology	A long-term average, giving a representative value of a quantity.
Light attenuation coefficient, Kd and Kd490	The light attenuation coefficient describes how rapidly light is attenuated as it passes through water: a high attenuation coefficient means that light is absorbed or scattered quickly by particles or coloured material in the water and light cannot penetrate very far, while low attenuation means that the water is clear. Attenuation varies with the wavelength of the light: Kd490 refers specifically to attenuation of light with wavelength 490 nm, in the middle of the visible range, while Kd refers to average attenuation for all wavelengths. Kd is inversely proportional to the Secchi depth.

RELEVANT DOCUMENT LIST

	Ref	Title	Version
DA1	CMEMS-NWS-PUM-004-002	Product User Manual for NWSHELF_ANALYSISFORECAST_BGC_LR_004_002	3.0
DA2	CMEMS-NWS-PUM-004-001	Product User Manual for NWSHELF_ANALYSISFORECAST_PHY_LR_004_001	1.0
DA3	CMEMS-NWS-QUID-004-001	Quality Information Document for NWSHELF_ANALYSISFORECAST_PHY_LR_004_001	1.0

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

NWSHELF_ANALYSISFORECAST_BGC_004_002: the biogeochemical part of a North West Shelf analysis and forecast performed at 7 km resolution.

Table 1: List of NWS-NRT BGC variables

Variable	Vertical resolution	Horizontal resolution	Temporal resolution
Chlorophyll-a	All levels, surface to bottom	7 km	daily average
Nitrate	All levels, surface to bottom	7 km	daily average
Phosphate	All levels, surface to bottom	7 km	daily average
Silicate	All levels, surface to bottom	7 km	daily average
Ammonium	All levels, surface to bottom	7 km	daily average
Oxygen	All levels, surface to bottom	7 km	daily average
Partial pressure of CO_2 (pCO_2)	Surface	7 km	daily average
pH	All levels, surface to bottom	7 km	daily average
Dissolved inorganic carbon (DIC)	All levels, surface to bottom	7 km	daily average
Total alkalinity	All levels, surface to bottom	7 km	daily average
Light attenuation coefficient (Kd)	All levels, surface to bottom	7 km	daily average
Depth of the euphotic zone	N/A	7 km	daily average
Phytoplankton biomass	All levels, surface to bottom	7 km	daily average
Net primary production	All levels, surface to bottom	7 km	daily average

I.2 Summary of the results

The quality of the product NWSHELF_ANALYSISFORECAST_BGC_004_002, the North West Shelf analysis and forecast system at 7 km resolution, has been assessed by comparison to observations using a two-year test run (2022-2023). Observational datasets used for comparison include satellite ocean colour chlorophyll and light attenuation, the North Sea Biogeochemistry Climatology, which is based on in situ measurements, gridded measurements from the Surface Ocean CO_2 Atlas (SOCAT) and Global Ocean

Data Analysis Project (GLODAP) and the International Council for the Exploration of the Sea (ICES) database (ICES, 2025) of in situ measurements for 2022-2023 (see below for references, section IV). The following paragraphs summarise the results of the validation for each variable.

Chlorophyll: temporal and spatial patterns of chlorophyll distribution at all depths are well reproduced. The model tends to overestimate chlorophyll during the growing season, especially in the more northern areas, though differences are mainly less than 1 mg m^{-3}. Regionally-averaged time series show that model values generally match the satellite values well, though they show lower variability.

Nitrate, phosphate and silicate: temporal and spatial patterns of surface nutrients distribution match those in a climatology of in situ data. The concentration values are generally in line with in situ data, although nitrate is overestimated, particularly in the Southern North Sea; for phosphate and silicate the direction of the bias varies with season. Some statistical comparisons have been made, but these should be used with caution, given the relatively low number of observations available, and it is not recommended to use them as Estimated Accuracy Numbers.

Ammonium: the assessment has high uncertainty and users should treat the values with caution.

Dissolved oxygen: Seasonal and spatial variation in surface oxygen values are captured well, though model values are lower than observed in the winter and higher for most areas in the growing season. Sub-surface values are higher than observation by 10-20 mmol m^{-3}, when averaged over a region, with deeper waters tending to have a larger bias.

Partial pressure of CO_2 at the surface: seasonal patterns of pCO_2 broadly match those seen in in situ data, with higher values in winter and in coastal areas. Summer values are overestimated compared to in situ observations.

pH, dissolved inorganic carbon and total alkalinity: the spatial distribution and size of the surface values are in line with a global climatology.

Light attenuation coefficient: values are higher than in a comparable satellite product (Kd490), but shows the same spatial and temporal patterns.

I.3 Estimated Accuracy Numbers

Estimated Accuracy Numbers (EANs) can only be produced with confidence for a limited number of variables due to lack of available observation data for biogeochemical variables. Biases are based on model minus observed values (i.e., positive when model values are larger) and root mean square difference (RMSD) is used as a measure of accuracy of the model according to the observations.

Surface log-chlorophyll accuracy, based on the match to daily ocean colour satellite values, is shown in Table 1.

Table1 Estimated accuracy numbers for surface log10-chloropyll, chlorophyll units mg m^{-3}. RMSD: Root Mean Square Difference; Bias: model – observation.

Region	**Bias** (mg m^{-3})	**RMSD** (mg m^{-3})
Continental shelf	-0.03	0.4
Full domain	0.005	0.4

The ocean colour data is taken from the CMEMS catalogue, product OCEANCOLOUR_ATL_BGC_L3_MY_009_113.

Table 2 shows estimated mean bias (model-observation) and root mean square difference (RMSD) for the on-shelf part of the model domain, based on comparison to matched in situ observations. **The estimates should be used with caution**: the in situ measurements are not distributed evenly in time or space and their accuracy is not known. Users are advised to consult the tables in section IV for more information about accuracy within each region: there is considerable variation between regions. EANs are given only for chlorophyll and oxygen, because there was insufficient in situ data available for other products.

Table 2 Estimated accuracy numbers for NWS BIO variables. Bias: model – observation. Values apply to the continental shelf region rather than to deeper waters.

Variable	**Mean bias**	**RMSD**	**Units**	**Decimal places**
Chlorophyll-a	-0.2	3.5	mg m^{-3}	1
Dissolved oxygen	14	31	mmol m^{-3}	0

II PRODUCTION SYSTEM DESCRIPTION

Production centres name:

North West Shelf MFC - Met Office, UK

Production system name:

Met Office – FOAM Shelf Seas Atlantic Margin Model 7 km (AMM7)

Description

The North West Shelf analysis and forecast is produced using the Forecasting Ocean Assimilation Model 7 km Atlantic Margin model (FOAM AMM7) which is comprised of version 3.6 of the Nucleus for European Modelling of the Ocean (NEMO) ocean model code (Madec et al., 2017) coupled to the European Regional Seas Ecosystem Model[1] (ERSEM, 22.11; Butenschön et al., 2016). The system incorporates data assimilation of surface chlorophyll from the CMEMS product OCEANCOLOUR_GLO_BGC_L3_NRT_009_101.

The model domain is located on the European North West continental Shelf (NWS), from 40°N, 20°W to 65°N, 13°E (see Figure 1), on a regular lat-lon grid with 1/15° latitudinal resolution and 1/9° longitudinal resolution (approximately 7 km square). The model domain is shown in Figure 1. Although the focus of this system is on the shallow seas of the shelf, the domain extends beyond the shelf to include some regions of deep water of the adjacent North-East Atlantic to ensure there is appropriate cross-shelf exchange. In the FOAM AMM7 system a hybrid s-sigma terrain-following coordinate system (following Siddorn and Furner, 2013) with 51 levels is employed in order to retain vertical resolution on the shelf. The NWS products for the biogeochemical and physical forecasts are however delivered on 24 geopotential (z-level) vertical levels based upon the ICES standard depths for ease of use and analysis. In addition, in the NWS products, model grid points near the model boundaries are strongly affected by the boundary conditions. Therefore, the outermost 10 grid points are masked at the open ocean boundary as well as the model grid points south of 57.25°N at the Baltic boundary.

[1] https://github.com/pmlmodelling/ersem/releases/tag/22.11)

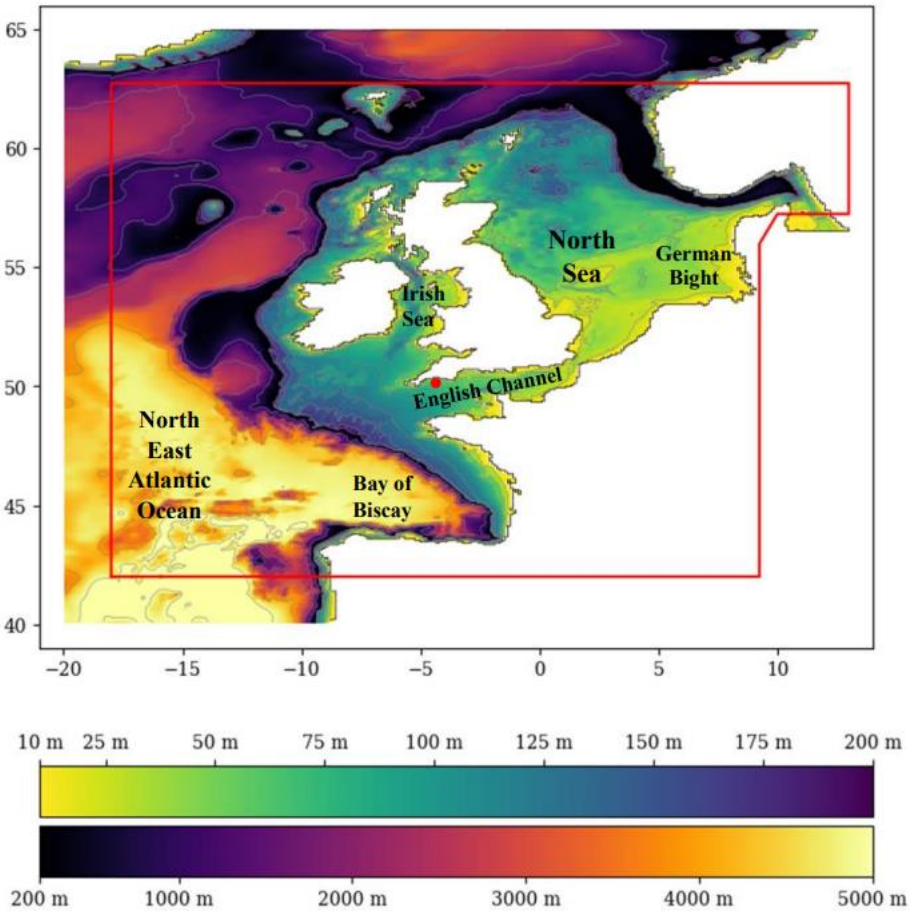

Figure 1: NWS bathymetry (m) of the AMM7 model domain, with the red line showing the product domain. The red dot shows the location of observing station L4 in the western English Channel. Model domain: European North West continental Shelf (NWS), from 40°N, 20°W to 65°N, 13°E. White areas show land, coloured areas show sea.

The following paragraphs describe the ERSEM biogeochemical model and its implementation in the FOAM AMM7 system. Physical variables from this modelling system are provided in the product NWSHELF_ANALYSISFORECAST_PHY_LR_004_001. For full details of that system, its forcing data and product quality please see its Quality Information Document (CMEMS-NWS-QUID-004-001-v1.0.pdf). ERSEM is coupled to the NEMO physical model through the Framework for Aquatic Biogeochemical Models (FABM, Bruggeman and Bolding, 2014). The ecosystem model is forced by the physical model via an online coupling and is run at the same time step as the physical model (300s).

ERSEM is a marine biogeochemistry model which simultaneously describes pelagic and benthic ecosystems in terms of phytoplankton, bacteria, zooplankton, zoobenthos and the biogeochemical cycling of C, N, P and Si. ERSEM uses a functional group approach to describe the ecosystem, whereby biota are grouped together according to their trophic level (subdivided according to first size, then trophic role and finally feeding method). Four functional groups of phytoplankton are included, three of zooplankton and one of bacteria. The dynamics of biological functional groups are described by processes relating to both physiology (ingestion, respiration, excretion and egestion) and population

(growth and mortality). The differences between the functional groups mainly lie in the rate constants, which are derived from the literature or from allometric considerations and in the food components on the uptake side. The carbonate system is included, enabling pCO_2, pH, dissolved inorganic carbon and total alkalinity to be included in the available products.

Full information on ERSEM can be found in Butenschön et al., (2016) and references therein. The model parameters match those given by Butenschön et al., (2016). Version 22.11 has a small number of changes compared to the 15.06 version described by Butenschön et al., (2016): pH is now provided on total scale instead of seawater scale, there are changes to handling of particulate matter at the sea bed and the formulations for air-sea exchange of oxygen and carbon dioxide have been harmonised and extended. Pelagic denitrification is now included, but in practice this does not occur in NWS conditions.

Production of the phytoplankton groups in ERSEM is driven by broadband photosynthetically active radiation (PAR), and the underwater light field is computed with the Beer-Lambert equation of light attenuation. The diffuse attenuation coefficient is computed by using broadband absorption and backscattering (i.e. the "Inherent Optical Properties") of the phytoplankton groups, of water, and of coloured detrital matter and other non-modelled material; the latter is nudged towards climatological satellite data. For this version of the NWS product, an updated version of the satellite data is used based on the ESA-CCI version 6.0 (Sathyendranath et al., 2019, 2023). Suspended particulate matter is not explicitly included in the model.

The NWS modelling system uses climatologies for the biogeochemical boundary conditions and river inputs. At the open boundary, values of nitrate, phosphate, silicate and oxygen are from World Ocean Atlas 2023 (Reagan et al., 2023) and alkalinity and dissolved inorganic carbon from GLODAPv2.2023 (Lauvset et al., 2024; Key et al., 2015; Lauvset at al., 2016). At the Baltic boundary fixed climatological concentration values are used for nutrients, while DIC and alkalinity are calculated from salinity. River nutrient concentrations are taken from a daily climatology compiled for the purpose. Nitrogen deposition at the surface uses the last available year from the EMEP project[2].

Satellite observations of surface chlorophyll are assimilated, using the NEMOVAR (Mogensen et al., 2009, 2012; Waters et al., 2015, King et al., 2018) assimilation scheme in its 3DVAR configuration. The method is as described in Skákala et al., (2018) for total surface chlorophyll concentration, except for minor differences in the treatment of observation error and correlation length scales used. The current version of the modelling system has adopted a recent set of error covariances and length scales from Fowler et al. (2023). The data used in the assimilation is total chlorophyll concentrations from Copernicus Marine TAC product OCEANCOLOUR_GLO_BGC_L3_NRT_009_101, specifically dataset cmems_obs-oc_glo_bgc-plankton_nrt_l3-multi-4km_P1D. A first guess at appropriate time technique is used to produce the differences between observations and the model. Model matchups are produced with bilinear interpolation to observation locations at midday. NEMOVAR calculates increments to total chlorophyll at the surface. Increments are then applied throughout the mixed layer, using an incremental analysis update (Bloom et al., 1996), to all four model phytoplankton variables by adjusting the total chlorophyll-a but maintaining the existing chlorophyll-a ratios at each model grid point between phytoplankton functional types. For each functional type, the carbon, nitrate, phosphate, and silicate contents are then adjusted according to the adjustment to their chlorophyll-a following

[2] http://www.emep.int

stoichiometry. In this version of the system the assimilation scheme has been modified so that in shallower waters, less than 60 m deep, only chlorophyll is updated and no increment is applied to C, N, P or Si. This is to avoid unrealistic accumulation of nitrate and phosphate in the southern North Sea and parts of the Irish Sea, which has occurred in previous versions.

III VALIDATION FRAMEWORK

The products assessed are chlorophyll, nitrate, phosphate, silicate, ammonium, oxygen, pCO_2, pH, total alkalinity, dissolved inorganic carbon and light attenuation. Assessment was made by comparison to observations using a two-year test run (2022-2023).

Model surface chlorophyll is compared to daily satellite ocean colour measurements (Copernicus Marine Service Ocean Colour TAC products OCEANCOLOUR_ATL_BGC_L3_MY_009_113 and OCEANCOLOUR_ATL_BGC_L4_MY_009_118) to give seasonal maps for the whole domain and time series plots and statistics for sub-regions.

Model nutrients, chlorophyll and oxygen are compared to climatological fields from the North Sea Biogeochemical Climatology (Hinrichs et al., 2017) to give seasonal maps of surface values.
Time series of surface nutrients are compared to in situ measurements from the Western Channel Observatory[3] (point L4; 50°15.0'N, 4°13.0'W).

Surface pCO_2 is compared to seasonal averages of the SOCATv2024 gridded coastal fields for 1990-2022, taken from CMEMS product INSITU_GLO_BGC_CARBON_DISCRETE_MY_013_050.
Surface pH, DIC and total alkalinity are compared to the annual average GLODAPv2.2023 gridded field, taken from CMEMS product INSITU_GLO_BGC_CARBON_DISCRETE_MY_013_050.

The light attenuation coefficient is compared to monthly satellite ocean colour measurements of Kd490 (OCEANCOLOUR_GLO_BGC_L4_MY_009_104) to give seasonal maps for the whole domain.

Nutrients, chlorophyll and oxygen are compared to in situ measurements at all depths from the database of the International Council for the Exploration of the Sea (ICES, 2025). This comparison has been used to produce Estimated Accuracy Numbers for chlorophyll and dissolved oxygen, but for other variables there are too few observations available for a quantitative assessment to be made.

The quantitative metrics provided in this document are defined below.

Number of matched observations (n) All analysis is based on the set of observations and a set of model outputs matched in time and space. In situ observations were matched to the daily mean model outputs: a match was considered valid if it was in the same grid cell, within 10 m vertically and 1 day of the observation. The number of matched points is shown in the tables in section IV and quantitative metrics are only given where the number and distribution of observations was sufficient to give a meaningful summary. This is considered to be more than about 200 points, though metrics based on this number of points should not be considered as a robust assessment for the full domain and all time of year.

Bias: mean of the difference between the model data and the observation (reference) data. This gives an indication of consistent differences between the model and observation, with positive bias meaning the model is higher than observation.

[3] www.westernchannelobservatory.org.uk. The Western Channel Observatory is funded by the UK Natural Environment Research Council through its National Capability Long-term Single Centre Science Programme, Atlantic Climate and Environment Strategic Science (AtlantiS), grant number NE/Y005589/1.

$$Bias = mean(x_{model} - x_{reference}) \quad \text{[Eq. 1]}$$

Root Mean Square Difference (RMSD): square root of the mean square difference between the model and observation values at each point. This indicates the average size of the difference between model and observation.

$$RMSD = \sqrt{\frac{\sum(x_{model} - x_{reference})^2}{n}} \quad \text{[Eq. 2]}$$

Correlation: Pearson correlation coefficient between the values of the model and observation data: if the model data increases when the observations do, they are positively correlated. 1 shows perfect correlation and 0 shows no correlation.

Statistics have been calculated for the whole model domain and for various subregions – see Figure 2 for the location of these regions.

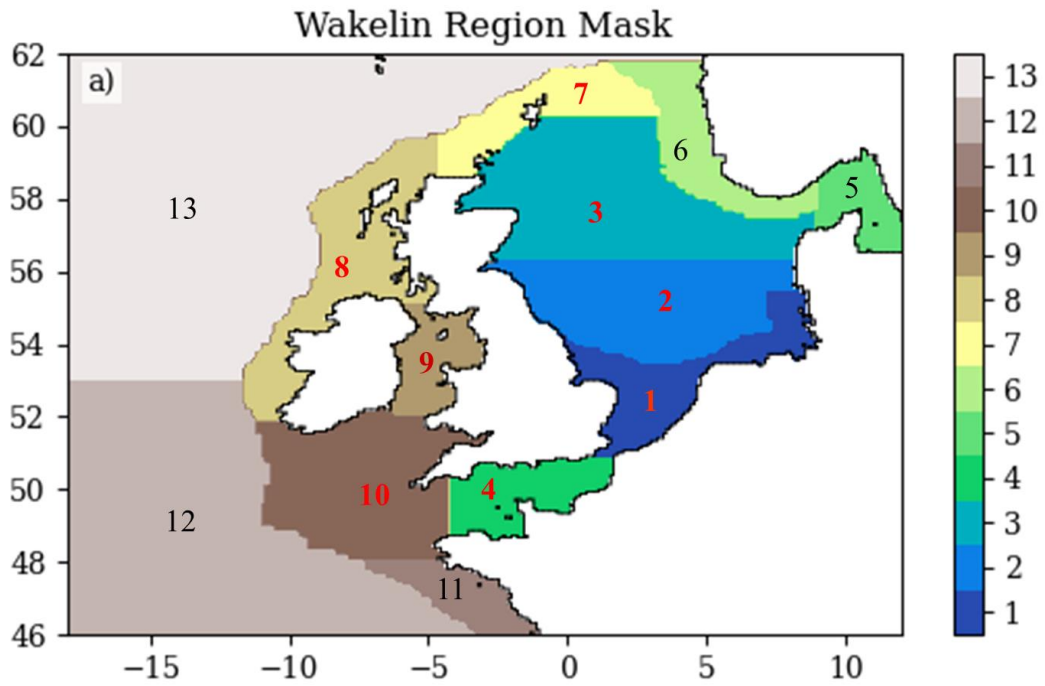

Figure 2: Regions considered during the validation, based on Wakelin et al. (2012). 1) Southern North Sea; 2) Central North Sea; 3) Northern North Sea; 4) English Channel; 5) Skagerrak Kattegat; 6) Norwegian Trench; 7) Shetland Shelf; 8) Irish Shelf; 9) Irish Sea; 10) Celtic Sea; 11) Armorican Shelf; 12) NE Atlantic S; 13) NE Atlantic N. References in the text to the Shelf region mean all except NE Atlantic N and S (12, 13), Armorican shelf (11), Norwegian Trench (6) and Skagerrak Kattegat (5). Shelf regions are labelled with red numbers, other regions are labelled with black numbers. White areas show land, coloured areas show sea.

IV VALIDATION RESULTS

In this section each variable is compared to available satellite and in situ observations, with one sub-section per variable. Results are presented as surface plots, enabling a visual comparison, and using statistical metrics where there are sufficient observations to support this.

IV.1 Chlorophyll-a

The chlorophyll-a product has been compared to values from satellite and in situ measurements for the test period (2022) that are detailed below. In Figure 3 the seasonal averages of surface chlorophyll are compared to satellite and in situ data. The satellite data is taken from the CMEMS level 4 monthly Globcolour product , and the climatology of in situ data is from the North Sea Biogeochemical Climatology (Hinrichs et al., 2017). Since the satellite data is not independent of the chlorophyll-a product, i.e., the level 3 data is assimilated and the level 4 data is used for validation, it is expected that the product resembles the satellite data. The in situ dataset is independent, and is also a time-averaged product, based on in situ data from 1970-2014, rather than representing a particular year.

The model reproduces the spatial and temporal variation seen in both observational datasets, with a clear seasonal cycle of low values in the winter and higher values in the spring and summer, and with shelf values generally higher than for the deeper ocean (Figure 3). The model tends to overestimate chlorophyll during the growing season, especially in the more northern areas, though differences are mainly less than 1 mg m-3. The exception is shallower waters near the coasts and in river plumes, where model values are lower than observed. However, we can be less confident of the difference here, as satellite measurements have higher uncertainty in these types of water and the in situ climatology is based on in situ data from 1970-2014, when river nutrient loads were higher.

Comparison to satellite observations is also presented using regionally-averaged time series (Figure 4) and statistics (Table 3), for a selection of the larger regions shown in Figure 2. To produce these, daily satellite level 3 data were interpolated on to the model grid and then matched to the model daily values. Data points missing due to cloud were excluded. Model values generally match the satellite values well, though they show lower variability (Figure 4). Table 3 gives statistics for log10-chlorophyll, which has a more normal distribution than chlorophyll so is less affected by extreme values. The bias is less than 0.1 everywhere except the Southern North Sea and Irish Sea. As also seen in Figure 3, biases tend to be positive in deeper areas (i.e. model higher than observations) and negative in shallower areas. Spatial correlation is about 0.6 for most regions, though somewhat lower for the Celtic Sea and the Southern North Sea (Table 3).

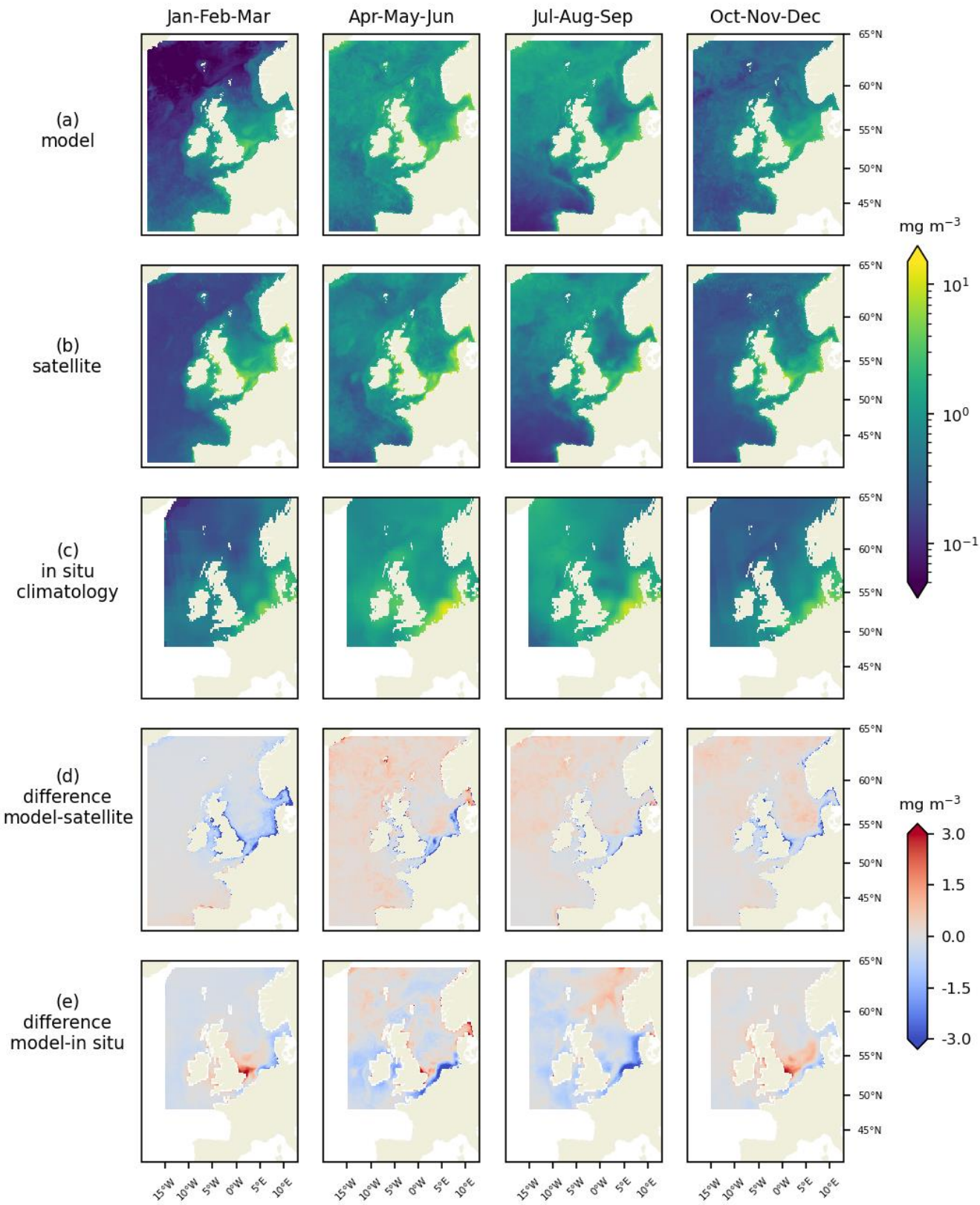

Figure 3: Seasonal average surface chlorophyll-a from (a) the model, 2022-23, (b) ocean colour satellite, CMEMS level 4 product 009_118, 2022 (c) climatology from the North Sea Biogeochemical Climatology in situ dataset (d) the difference between model and satellite values (e) the difference between model and in situ values.

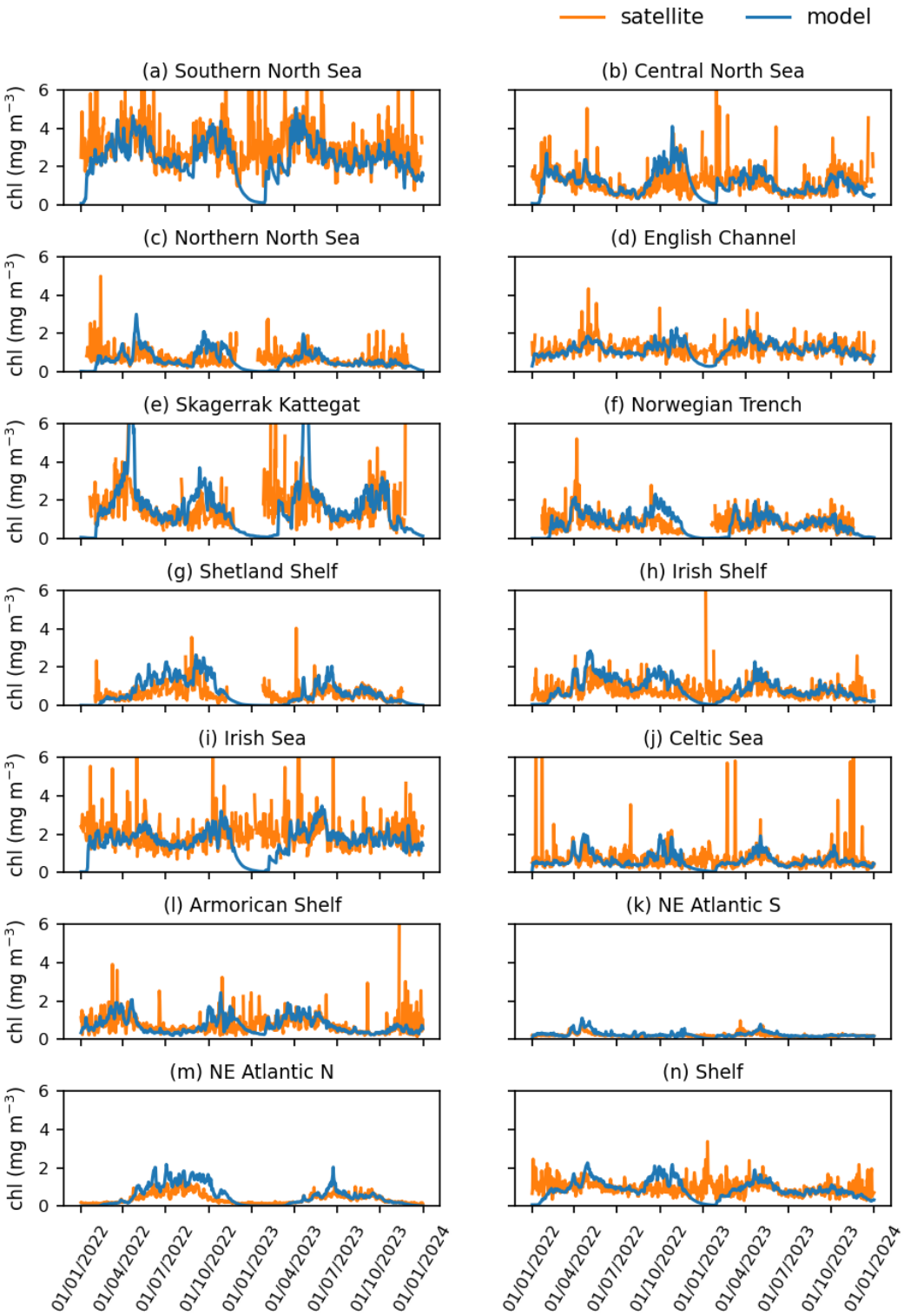

Figure 4 Time series of daily median chlorophyll for regions of the North West shelf area, 2022-23, from the model (blue line) and from ocean colour satellite, CMEMS product 009_118 (orange line). See Figure 2 for the location of the regions.

Table 3: Statistics for match-ups between model surface log-chlorophyll outputs and satellite ocean colour chlorophyll for the full domain and selected larger sub-regions, using daily fields for 2022-23. The chlorophyll unit is mg m^{-3}. Bias: model-observations. RMSD: Root Mean Square Difference. Correlation: Pearson correlation coefficient. See Figure 2 for the location of the regions

Region	Bias (mg m^{-3})	RMSD (mg m^{-3})	Correlation
Full Domain	0.005	0.413	0.71
Shelf	-0.031	0.360	0.66
Northern North Sea	-0.010	0.375	0.59
Central North Sea	0.007	0.334	0.60
Southern North Sea	-0.137	0.398	0.37
English Channel	-0.041	0.220	0.66
Celtic Sea	-0.144	0.470	0.28
Irish Sea	-0.043	0.325	0.67
NE Atlantic S	0.050	0.340	0.63
NE Atlantic N	-0.007	0.560	0.64

Statistics from match-ups of model chlorophyll outputs at all depths to in situ observations from the ICES database are given in Table 4, for regions where there was enough observational data for a meaningful comparison. The bias values are averages for all the times when the observations were made and therefore are not distributed evenly through the year: they should be used with caution.

Table 4 Statistics for match-ups between daily model chlorophyll outputs for 2022-2023 and in situ observations at all depths from the ICES database, for selected sub-regions. Bias: model-observations. RMSD: Root Mean Square Difference. Correlation: Pearson correlation coefficient. See Figure 2 for the location of the regions.

Region	number of points	Bias (mg m^{-3})	RMSD (mg m^{-3})	Correlation
Shelf	2121	-0.17	3.52	0.45
Northern North Sea	315	-0.39	3.72	0.33
Central North Sea	373	0.23	1.52	0.65
Southern North Sea	806	-1.20	4.61	0.41
Celtic Sea	212	1.50	2.51	-0.01
Irish Sea	256	0.48	2.43	0.45

IV.2 Nitrate

The comparison between the seasonal averages of surface nitrate and the climatology of in situ data from the North Sea Biogeochemical Climatology (Hinrichs et al., 2017) are shown in Figure 5. The model generally reproduces well the spatial and temporal patterns seen in the climatology, though there is a general tendency to higher values than observed. The seasonal cycle of nutrient depletion in the spring and summer and restoration in the winter is reproduced well. The biggest differences are seen in the Southern North Sea, where the model overestimates nitrate compared to the observational climatology.

The North Sea Biogeochemical Climatology is based on in situ data from 1970-2014 and river nutrient loads have decreased over that period so the in situ values may be overestimated in areas affected by river inputs, such as the German Bight (Figure 1).

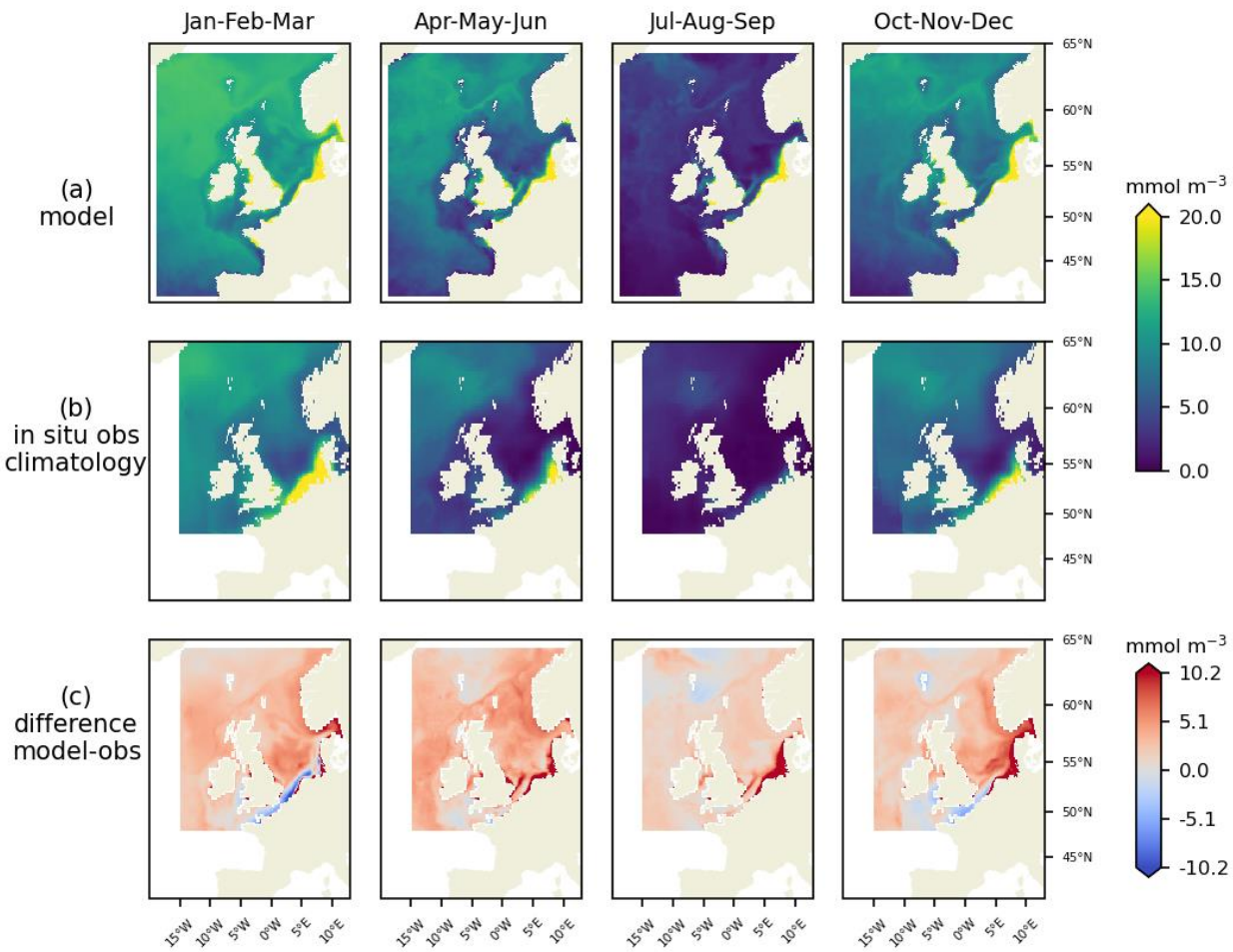

Figure 5 Seasonal average surface nitrate from (a) the model, 2022-23, (b) in situ climatology from the North Sea Biogeochemical Climatology dataset, (c) the difference between model and in situ values.

Figure 6 shows a time series of surface nitrate from the model against in situ data from station L4 in the western English Channel. The seasonal pattern of nitrate variation is broadly captured by the model, with high winter values of ~7 mmol m^{-3} dropping quickly to low values in April-May. However, the autumn rise occurs earlier in the model than in the in situ measurements and the spring fall is a little delayed. The model overestimates nitrate for the second half of the year.

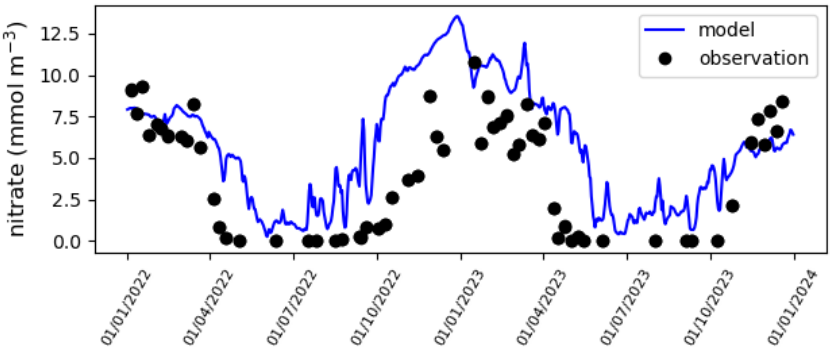

Figure 6: Time series of surface nitrate at location L4 (station in the western English Channel; see Figure 1 for the location), 2022-23; model (blue solid line) and in situ observation (black dots).

Table 5 shows statistics from match-ups of model nitrate outputs to observations at all depths from the ICES database, for regions where there was enough observational data for a meaningful comparison. Consistent with the surface plots in Figure 5, the model values are higher than observed, especially in the Southern North Sea. However, the correlations are reasonably good (≥0.5 for most regions), giving some evidence that the model reproduces spatial and temporal patterns at all depths. These statistics should be used with caution, given the relatively low number of observations available, and it is not recommended to use them as Estimated Accuracy Numbers, but they may be helpful for giving some information about the variation in accuracy between regions.

Table 5: Statistics for match-ups between daily model nitrate outputs for 2022-2023 and in situ observations at all depths from the ICES database, for selected sub-regions. Bias: model-observations. RMSD: Root Mean Square Difference. Correlation: Pearson correlation coefficient. See Figure 2 for the location of the regions.

Region	**number of points**	**Bias** (mg m^{-3})	**RMSD** (mg m^{-3})	**Correlation**
Shelf	1844	4.7	9.1	0.51
Northern North Sea	588	2.7	3.0	0.90
Central North Sea	109	3.7	4.6	0.78
Southern North Sea	225	17.8	24.2	0.43
Irish Sea	244	1.9	4.1	0.54

IV.3 Phosphate

The comparison between the seasonal averages of surface phosphate and the climatology of in situ data from the North Sea Biogeochemical Climatology (Hinrichs et al., 2017) are shown in Figure 9. The model reproduces the observed seasonal cycle of higher winter values and lower summer values, though with

some overestimation in the winter and underestimation in the summer. The exception is the continental coast of the Southern North Sea, where values are less than observed for most of the year. It should be noted that the North Sea Biogeochemical Climatology is based on in situ data from 1970-2014 and river nutrient loads have decreased over that period so the in situ values may be overestimated in areas affected by river inputs, such as the German Bight (see Figure 1 for location).

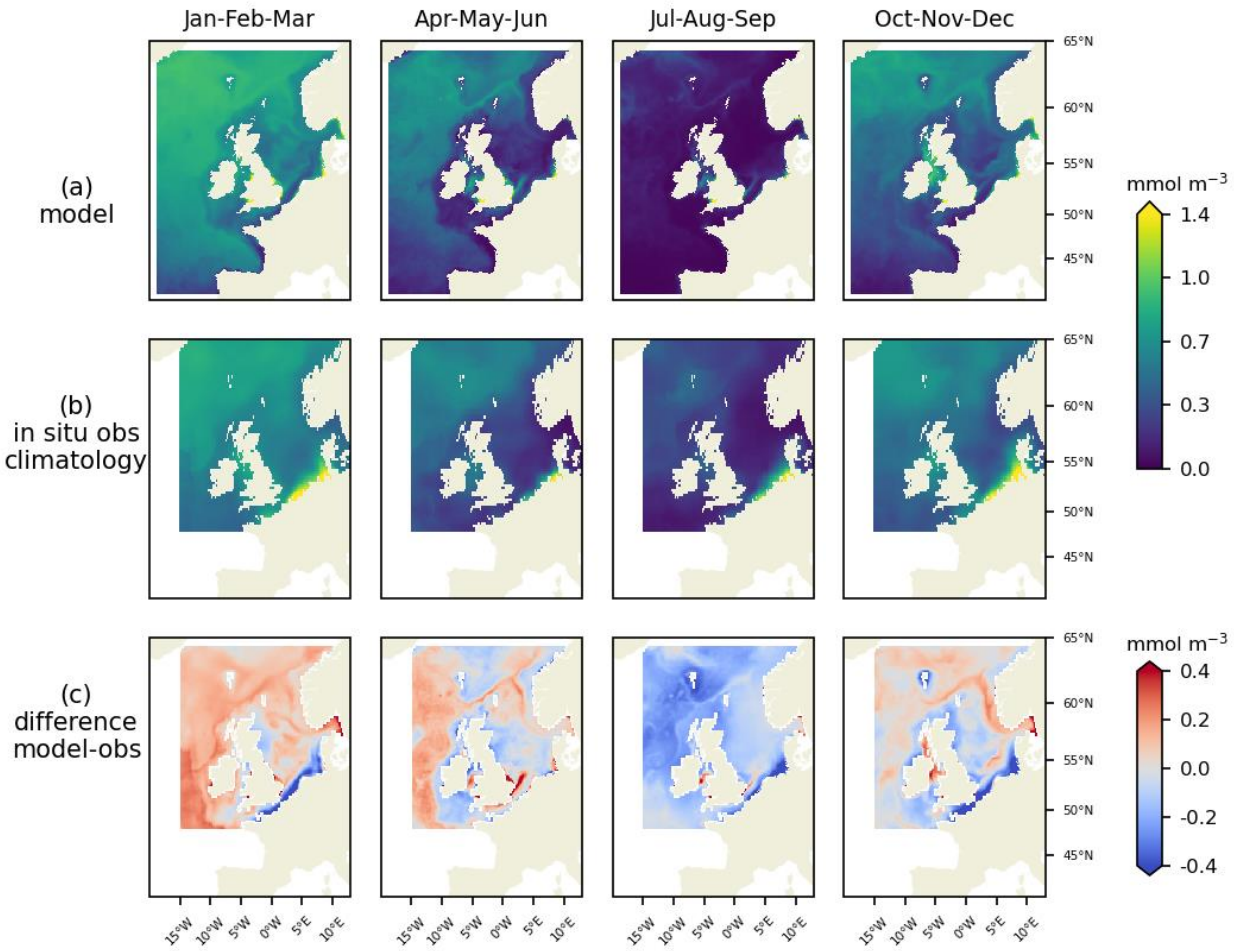

Figure 7: Seasonal average surface phosphate from (a) the model, 2022-23, (b) in situ climatology from the North Sea Biogeochemical Climatology dataset, (c) the difference between model and in situ values..

Figure 8 shows a comparison of model outputs of surface phosphate for 2022 to a time series of in situ data from station L4 in the western English Channel. The seasonal pattern is well captured by the model, with good agreement on timings of both the spring reduction in nutrients and the autumn increase. Winter values are overestimated in winter 2022/23 but underestimated at the end of 2023 suggesting that winter values are less reliable than summer, at least at this location.

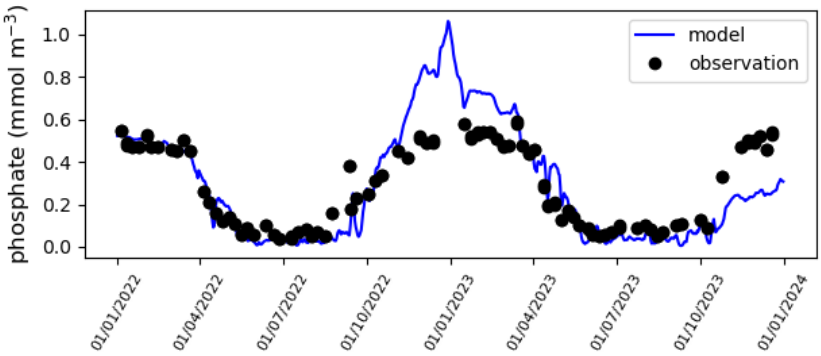

Figure 8: Time series of surface phosphate at location L4 (station in the western English Channel; see Figure 1 for the location), 2022-23; model (blue solid line) and in situ observation (black dots).

Table 6 shows statistics from match-ups of model phosphate outputs to observations at all depths from the ICES database, for regions where there was enough observational data for a meaningful comparison. The overall bias is small, though model values are higher than observed in the Southern North Sea – this contrasts with Figure 7, where the model is mainly lower than observations for this region, and may reflect changing river nutrient inputs that are not captured by the model. These statistics should be used with caution, given the relatively low number of observations available, and it is not recommended to use them as Estimated Accuracy Numbers, but they may be helpful for giving some information about the variation in accuracy between regions.

Table 6: Statistics for match-ups between daily model phosphate outputs for 2022-2023 and in situ observations at all depths from the ICES database, for selected sub-regions. Bias: model-observations. RMSD: Root Mean Square Difference. Correlation: Pearson correlation coefficient. See Figure 2 for the location of the regions.

Region	**number of points**	**Bias** (mg m^{-3})	**RMSD** (mg m^{-3})	**Correlation**
Shelf	3423	0.05	0.49	0.24
Northern North Sea	639	0.07	0.16	0.79
Central North Sea	216	0.03	0.16	0.80
Southern North Sea	647	0.13	0.42	0.33
Celtic Sea	398	-0.12	0.79	0.19
Irish Sea	659	-0.07	0.68	0.01

IV.4 Silicate

The comparison between the seasonal averages of surface silicate and the climatology of in situ data from the North Sea Biogeochemical Climatology (Hinrichs et al., 2017) are shown in Figure 9. The model reproduces the observed seasonal cycle of higher winter values and lower summer values, though with some tendency to overestimate in the winter and underestimate in the summer for deeper waters. For shallower coastal areas, especially the continental margin, the opposite is true, with model values lower than observed in the winter and lower in the summer. As for nitrate and phosphate, the largest absolute differences in the model occur along the coast of the German Bight (see Figure 1 for location) in all seasons. It should be noted that the North Sea Biogeochemical Climatology is based on in situ data from 1970-2014 and river nutrient loads have decreased over that period so the in situ values may be overestimated in areas affected by river inputs, such as the German Bight.

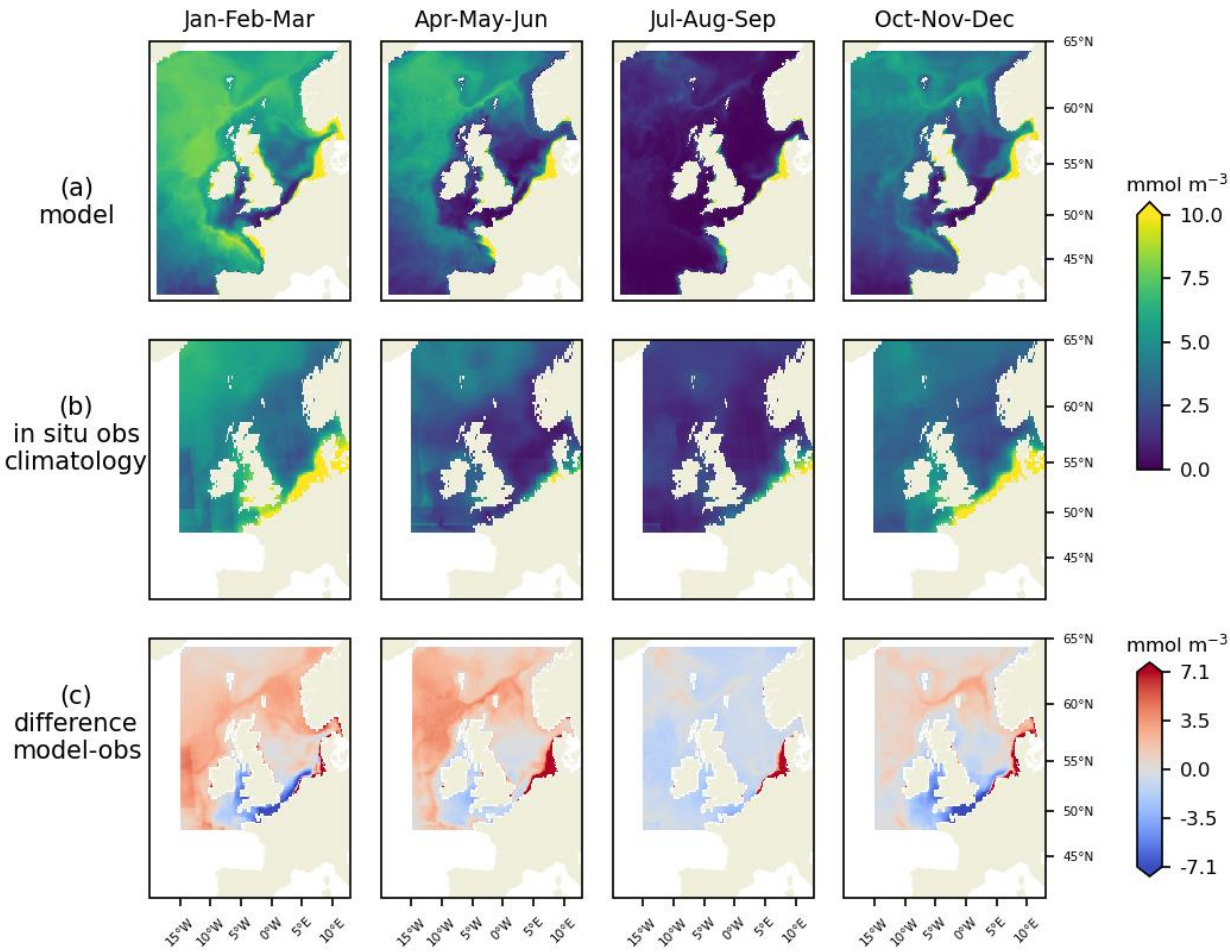

Figure 9: Seasonal average surface silicate from (a) the model, 2022-23, (b) in situ climatology from the North Sea Biogeochemical Climatology dataset, (c) the difference between model and in situ values.

Figure 10 shows a comparison of model outputs of surface silicate for 2022 to a time series of in situ data from station L4 in the western English Channel. The model simulates the seasonal cycle of higher

winter and lower summer values; in particular, the timing of the spring fall in silicate is modelled well. However, the autumn rise occurs later than observed and the winter values are higher.

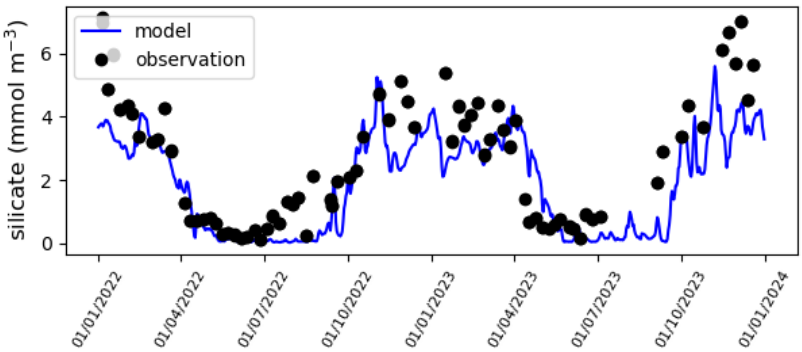

Figure 10: Time series of surface silicate at location L4 (station in the western English Channel; see Figure 1 for the location), 2022-23; model (blue solid line) and in situ observation (black dots).

Table 7 shows statistics from match-ups of daily model silicate outputs to observations at all depths from the ICES database, for regions where there was enough observational data for a meaningful comparison. These statistics should be used with caution, given the relatively low number of observations available, and it is not recommended to use them as Estimated Accuracy Numbers, but they may be helpful for giving some information about the variation in accuracy between regions.

Table 7: Statistics for match-ups between daily model silicate outputs for 2022-2023 and in situ observations at all depths from the ICES database, for selected sub-regions. Bias: model-observations. RMSD: Root Mean Square Difference. Correlation: Pearson correlation coefficient. See Figure 2 for the location of the regions.

Region	number of points	Bias (mg m^{-3})	RMSD (mg m^{-3})	Correlation
Shelf	3394	2.34	11.78	0.24
Northern North Sea	628	1.59	4.54	0.57
Central North Sea	216	5.77	11.65	0.63
Southern North Sea	648	12.01	20.86	0.32
Celtic Sea	398	-1.14	5.98	0.30
Irish Sea	637	-5.11	13.45	0.15

IV.5 Ammonium

The comparison between the seasonal averages of surface ammonium and the climatology of in situ data from the North Sea Biogeochemical Climatology (Hinrichs et al., 2017) are shown in Figure 11. The

model consistently underestimates ammonium values compared to the climatology, however this is not seen in comparison to observations at station L4 (Figure 12), where model values are higher than observed. Comparison to matched observational data (Table 8) shows a negative bias in most regions, and the RMSD is high in relation to mean values.

Given the high uncertainty in the assessment for ammonium users should treat the values with caution. They may be appropriate for some uses but they are not recommended for applications where high accuracy is required.

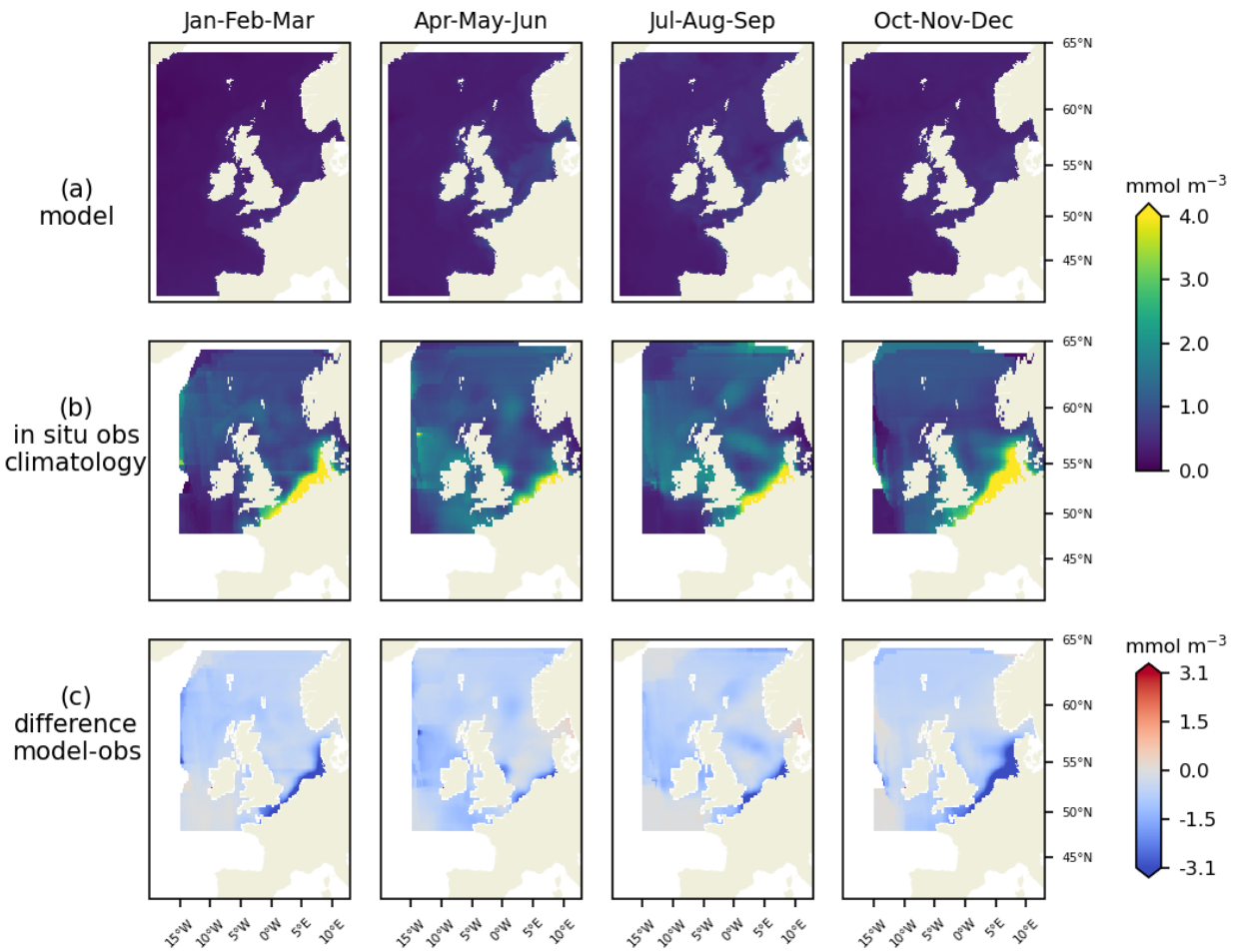

Figure 11 Seasonal average surface ammonium from (a) the model, 2022-23, (b) in situ climatology from the North Sea Biogeochemical Climatology dataset, (c) the difference between model and in situ values.

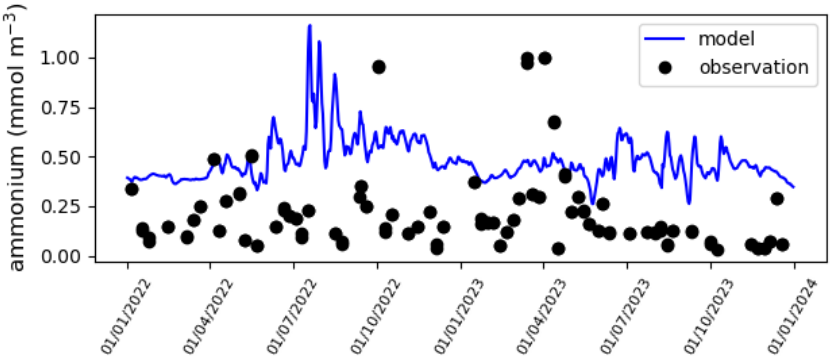

Figure 12 Time series of surface ammonium at location L4 (station in the western English Channel; see Figure 1 for the location), 2022-23; model (blue solid line) and in situ observation (black dots).

Table 8 Statistics for match-ups between daily model ammonium outputs for 2022-2023 and in situ observations at all depths from the ICES database, for selected sub-regions. Bias: model-observations. RMSD: Root Mean Square Difference. Correlation: Pearson correlation coefficient. See Figure 2 for the location of the regions.

Region	number of points	Bias (mg m^{-3})	RMSD (mg m^{-3})	Correlation
Shelf	1682	-1.45	5.56	0.14
Northern North Sea	152	0.10	0.66	0.17
Central North Sea	158	-0.20	1.14	0.23
Southern North Sea	647	-1.25	2.72	-0.07
Celtic Sea	277	-1.12	3.43	0.12
Irish Sea	307	-3.76	11.48	0.19

IV.6 Dissolved oxygen

The comparison between the seasonal averages of surface oxygen and the climatology of in situ data from the North Sea Biogeochemical Climatology (Hinrichs et al., 2017) are shown in Figure 13. Seasonal and spatial variation are captured well, though model values are lower than observed in the winter and higher for most areas in the growing season.

The North Sea Biogeochemical Climatology is based on in situ data from 1970-2014 and sea surface temperatures have risen during and since that period, leading to lower oxygen solubility in recent years. Therefore, it is likely that this climatology overestimates dissolved oxygen compared to present-day values.

Statistics from match-ups of model oxygen outputs at all depths to observations from the ICES database are given in Table 9, for regions where there was enough observational data for a meaningful

comparison. Biases are positive for all regions and tend to be larger for deeper water. Correlations are above 0.5 for most regions, showing that temporal and spatial patterns are well captured by the model.

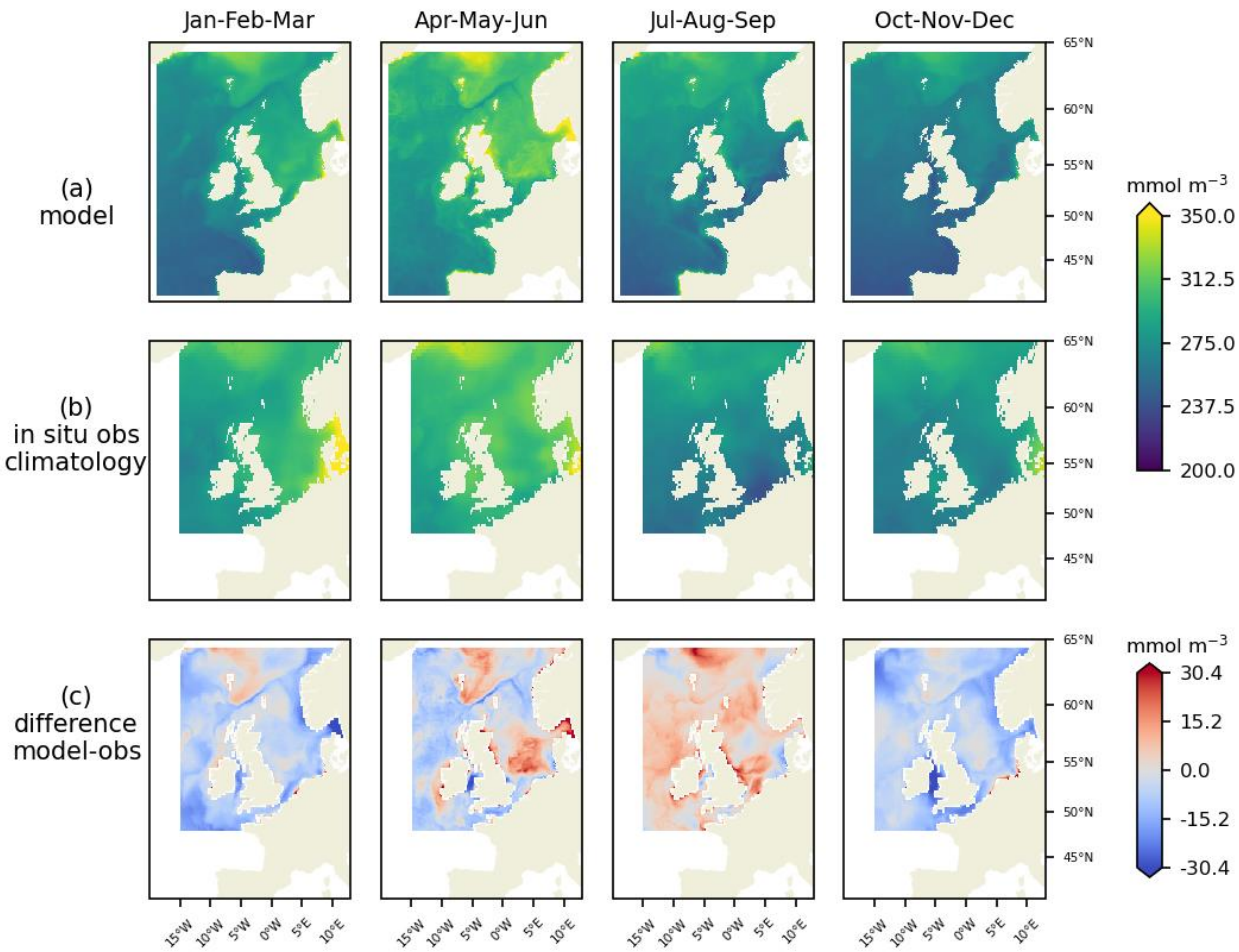

Figure 13: Seasonal average surface dissolved oxygen from (a) the model, 2022-23, (b) in situ climatology from the North Sea Biogeochemical Climatology dataset, (c) the difference between model and in situ values.

Table 9 Statistics for match-ups between daily model oxygen outputs for 2022-2023 and in situ observations at all depths from the ICES database, for selected sub-regions. Bias: model-observations. RMSD: Root Mean Square Difference. Correlation: Pearson correlation coefficient. See Figure 2 for the location of the regions.

Region	number of points	Bias (mg m^{-3})	RMSD (mg m^{-3})	Correlation
Shelf	65692	14.0	31.2	0.68
Northern North Sea	13681	13.3	30.4	0.65
Central North Sea	17153	15.0	30.5	0.72
Southern North Sea	6246	7.4	22.8	0.86

Region	number of points	Bias (mg m^{-3})	RMSD (mg m^{-3})	Correlation
English Channel	1281	8.0	9.7	0.45
Celtic Sea	15187	14.5	36.4	0.55
Irish Sea	4046	2.1	25.9	0.61
NE Atlantic S	25565	17.5	30.8	0.43
NE Atlantic N	24291	10.5	18.6	0.80

IV.7 Carbonate system variables: surface pCO_2, pH, total alkalinity and dissolved inorganic carbon (DIC)

ERSEM includes a model of the carbonate system. Dissolved inorganic carbon (DIC) and total alkalinity are state variables, with pH and the partial pressure of CO_2 (pCO_2) being calculated from them at each time step. All four of these variables are included in this section.

The modelled surface partial pressure of CO_2 (pCO_2) was compared to in situ data from the SOCAT v2024 database[6] (Bakker et al., 2016). The SOCAT values are for fugacity of CO_2, but the difference between fugacity and partial pressure is small in the conditions of this region.

Figure 14 shows seasonal mean fields from the model and seasonal averages of the monthly values in the SOCAT gridded coastal dataset. SOCAT values from years before 1990 were not used, to reduce the effect of the increase in atmospheric CO_2 concentrations during the observation period. The SOCAT dataset gives incomplete coverage, but the same broad patterns can be seen in the model and in situ data, with higher values in winter and in coastal areas. Summer values (July-September) appear higher in SOCAT than the model outputs.

[6] CMEMS product INSITU_GLO_BGC_CARBON_DISCRETE_MY_013_050

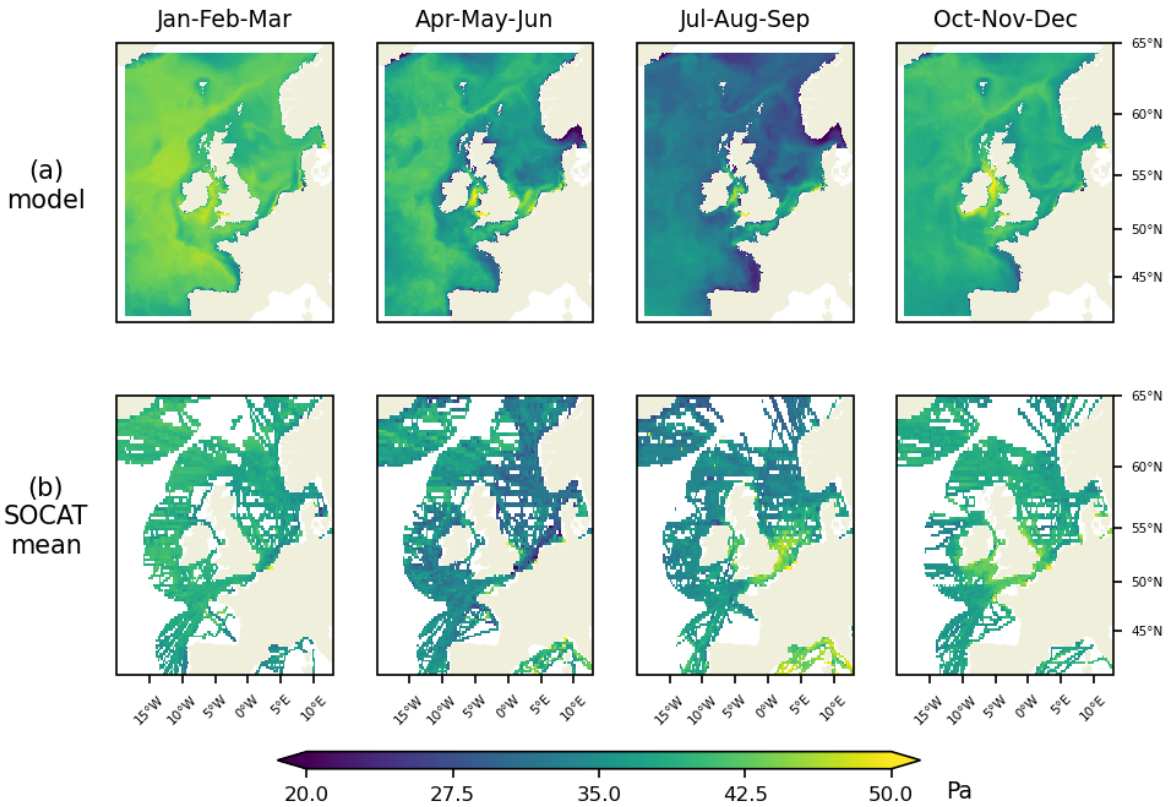

Figure 14: Seasonal average surface pCO2 from model, 2022-23, and gridded in situ observations (SOCAT, average for 1990-2022).

Figure 12 shows annual mean surface fields from the model and the GLODAPv2.2023[7] gridded fields for pH, DIC and total alkalinity. The spatial patterns and size of the modelled values broadly match GLODAP – for example, lower pH values in the southern North Sea and higher DIC in the north west – but the relatively low resolution of the GLODAP product limits the possible comparison. Relatively few observations are available for the shelf area, so a quantitative comparison has not been possible.

[7] CMEMS product INSITU_GLO_BGC_CARBON_DISCRETE_MY_013_050

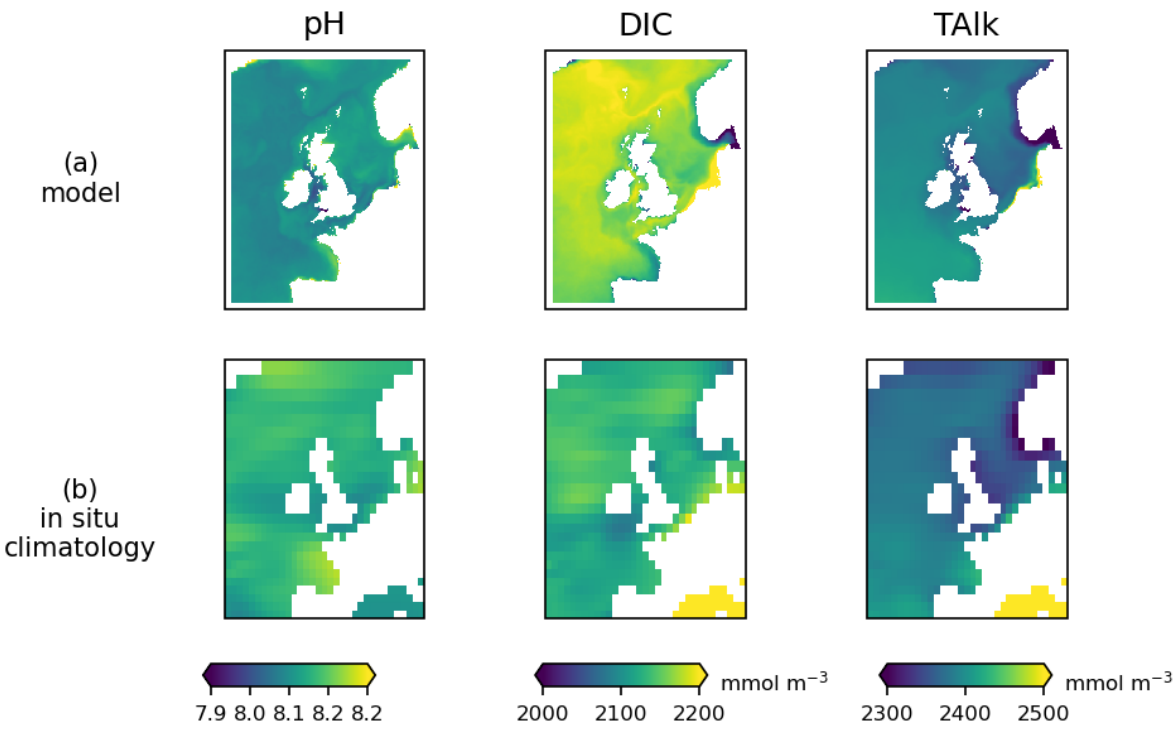

Figure 15: Annual average surface pH, dissolved inorganic carbon (DIC) and total alkalinity (TAlk) from the model, 2022-23, and gridded in situ climatology (GLODAP).

IV.8 Light attenuation Kd

The seasonal average attenuation coefficient at the surface for the model and for satellite-derived values for Kd490[8] are shown in Figure 16. Satellite Kd490 is a comparable but not identical measure of attenuation to the model Kd: Kd490 refers to attenuation at 490 nm while the model has no spectral dependence.

Model values are consistently higher than observations with the largest differences in the shallow waters of the Southern North Sea (the satellite values have highest uncertainty in this type of water). The bias is particularly high in the winter – this may be caused by the model's limited ability to capture light attenuation by inorganic sediment, which tends to be higher in the winter because of stronger wind mixing. The summer increase in attenuation, caused by the increase in phytoplankton and other organic material, is better captured by the model.

[8] CMEMS product OCEANCOLOUR_GLO_BGC_L4_MY_009_104

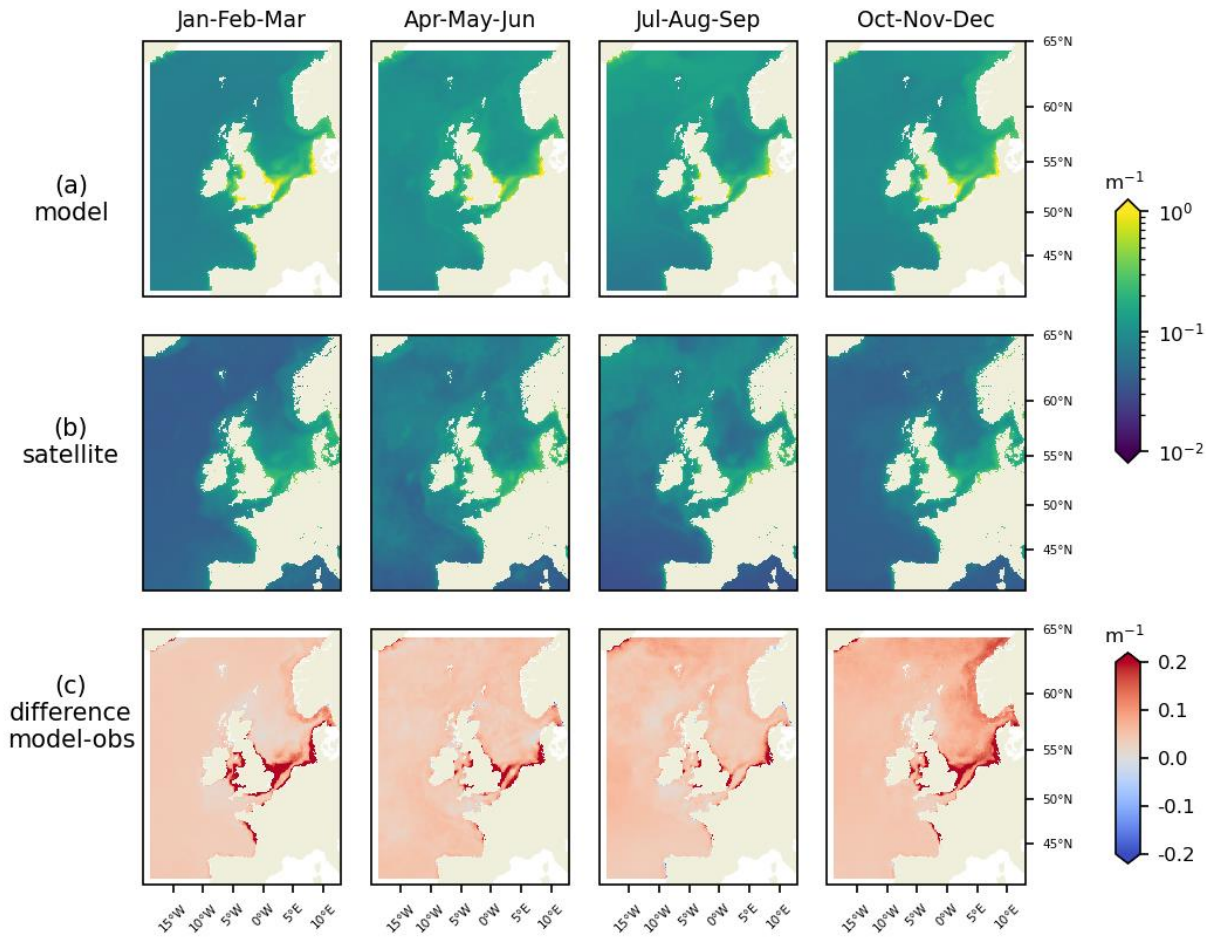

Figure 16: Seasonal average light attenuation coefficient from the model and satellite Kd490.

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to describe and track changes that may occur in the current operational system.

Date	Change/Event description	System version	Other
Nov 2025	Entry into Service, replacing previous service (new Production Unit).	AMM7	

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This modelling system is a new and updated version of the Met Office system which was used to provide the analysis and forecast product prior to November 2023. It has similar quality to that product, though with some small improvements. The product delivered from November 2023 to November 2025 used a different modelling system and the reader is advised to refer to its Quality Information Document for details. Broadly speaking, the two systems have comparable quality but the system described here has more accurate estimates of chlorophyll and pH, while the 2023-2025 system had more accurate estimates of nitrate and phosphate.

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